

SLIP – 11

Section I: Data Structures [15 Marks]

Q 1) Write a 'C' program to reverse a string using Static implementation of Stack.

Answer:

```
#include <stdio.h>
#include <string.h>
#define MAX 100

char stack[MAX]; int
top = -1;

void push(char c)
{
    if(top == MAX - 1)
        printf("Stack Overflow\n");    else
        stack[++top] = c;
}

char pop()
{
    if(top == -1)
        return '\0';
    return stack[top--];
}

int main()
{
    char str[MAX];
    int i;

    printf("Enter a string: ");
    scanf("%s", str);

    for(i = 0; str[i] != '\0'; i++)
        push(str[i]);

    printf("Reversed string: ");
    while(top != -1)
        printf("%c", pop());

    return 0;
}
```

Section II: Database Management Systems [15 Marks]

Q 2) Create the following tables (primary keys are underlined).

Emp (eno, ename, salary)

Project (pno, pname, budget)

Relationships: Emp– Project : Many to Many with descriptive attribute total_hours.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Emp(
    eno
    INT PRIMARY KEY,
    ename
    VARCHAR(50),
    salary
    NUMERIC(10,2)
);
```

```
CREATE TABLE Project(
    pno INT PRIMARY KEY,
    pname VARCHAR(50),
    budget NUMERIC(12,2)
);
```

```
CREATE TABLE Emp_Project(
    eno INT
    REFERENCES Emp(eno),
    pno INT
    REFERENCES Project(pno),
    total_hours INT,
    PRIMARY KEY(eno, pno)
);
```

```
INSERT INTO Emp VALUES
(1, 'Sneha', 30000),
(2, 'Rahul', 25000),
(3, 'Meena', 40000);
```

```
INSERT INTO Project VALUES
(10, 'Robotics', 500000), (20, 'Website', 200000);
```

```
INSERT INTO Emp_Project VALUES
(1, 10, 120),
(2, 20, 60);
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. Find the maximum budget.

Answer:

```
SELECT MAX(budget) FROM Project;
```

ii. Increase the salary of all employees by 10%.

Answer:

```
UPDATE Emp
SET salary = salary * 1.10;
```

iii. Count the number of projects having duration greater than 100 hrs.

Answer:

```
SELECT COUNT(*)
FROM
(
    SELECT pno
    FROM Emp_Project
    GROUP BY pno
    HAVING SUM(total_hours) > 100
) T;
```

iv. List all employees whose name ends with "a".

Q 2(C)

C. Write a query to list the names of employees who are not working on any project. [4 marks]

Answer:

```
SELECT ename
FROM Emp E
WHERE NOT EXISTS
(
    SELECT 1 FROM Emp_Project EP
    WHERE EP.eno = E.eno
)
```

) ;

Q 3) Viva [5 marks]